

CURING OF CONCRETE

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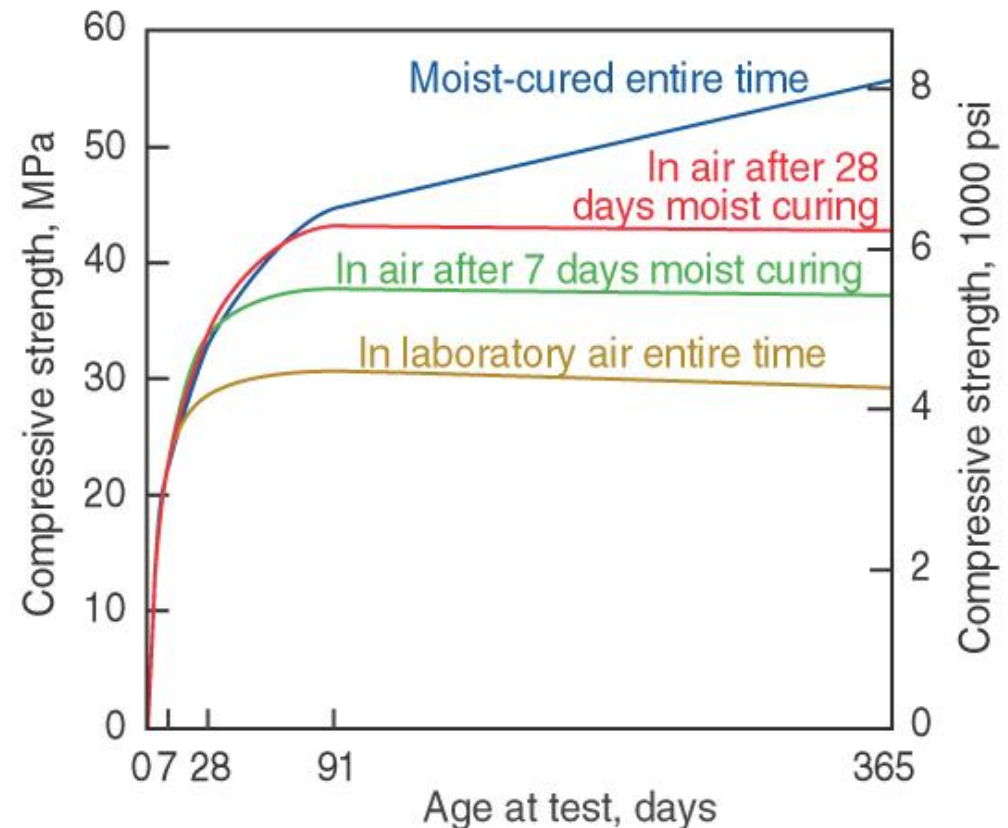
CURING OF CONCRETE

- Properties of concrete can improve with age as long as conditions are favorable for the continued hydration of cement. These improvements are rapid at early ages and continues slowly for an indefinite period of time.
- Curing is the procedures used for promoting the hydration of cement and consists of a control of temperature and the moisture movement from and into the concrete.

CURING OF CONCRETE

- The primary objective of curing is to keep concrete saturated or as nearly saturated as possible.

- Hydration reactions can take place in only saturated water filled capillaries.



Curing Methods

- Methods which supply additional water to the surface of concrete during early hardening stages.
 - Using wet covers
 - Sprinkling
 - Ponding



Curing Methods

- Methods that prevent loss of moisture from concrete by sealing the surface.
 - Water proof plastics
 - Use liquid membrane-forming compounds
 - Forms left in place



Curing Methods

- Methods that accelerate strength gain by supplying heat & moisture to the concrete.
 - By using live steam (steam curing)
 - Heating coils.



Hot Weather Concrete

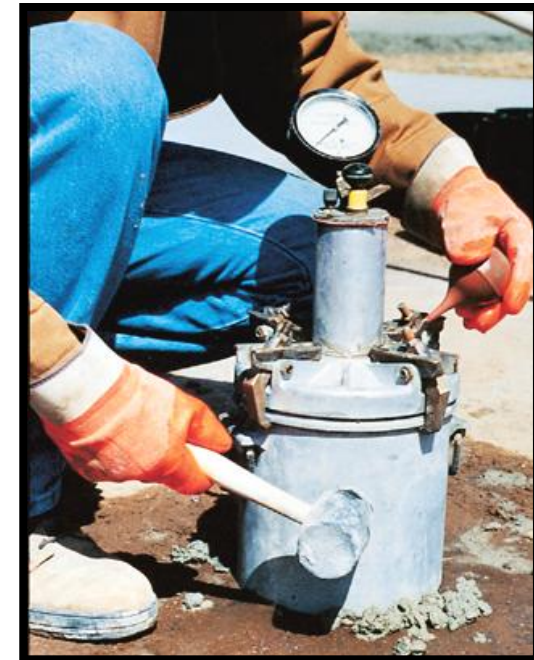
- Rapid hydration → early setting → rapid loss of workability
- Extra problems due to
 - Low humidity
 - Wind, excessive evaporation
 - Direct sunlight
- Solutions
 - Windbreaks
 - Cooled Concrete Ingredients
 - Water ponding (cooling due to evaporation)
 - Reflective coatings/coverings

Cold Weather Concrete

- Keep concrete temperature above 5 °C to minimize danger of freezing
- Solutions
 - Heated enclosures, insulation
 - Rely on heat of hydration for larger sections
 - Heated ingredients --- concrete hot when placed
 - High early strength cement

UNIFORMITY OF CONCRETE

- Concrete uniformity is checked by conducting tests on fresh and hardened concretes.
 - Slump, unit weight, air content tests
 - Strength tests



UNIFORMITY OF CONCRETE

- Due to heterogeneous nature of concrete, there will always be some variations. These variations are grouped as:
 - Within-Batch Variations : inadequate mixing, non-homogeneous nature
 - Batch-to-Batch Variations : type of materials used, changes in gradation of aggregates, changes in moisture content of aggregates

PROPERTIES OF HARDENED CONCRETE

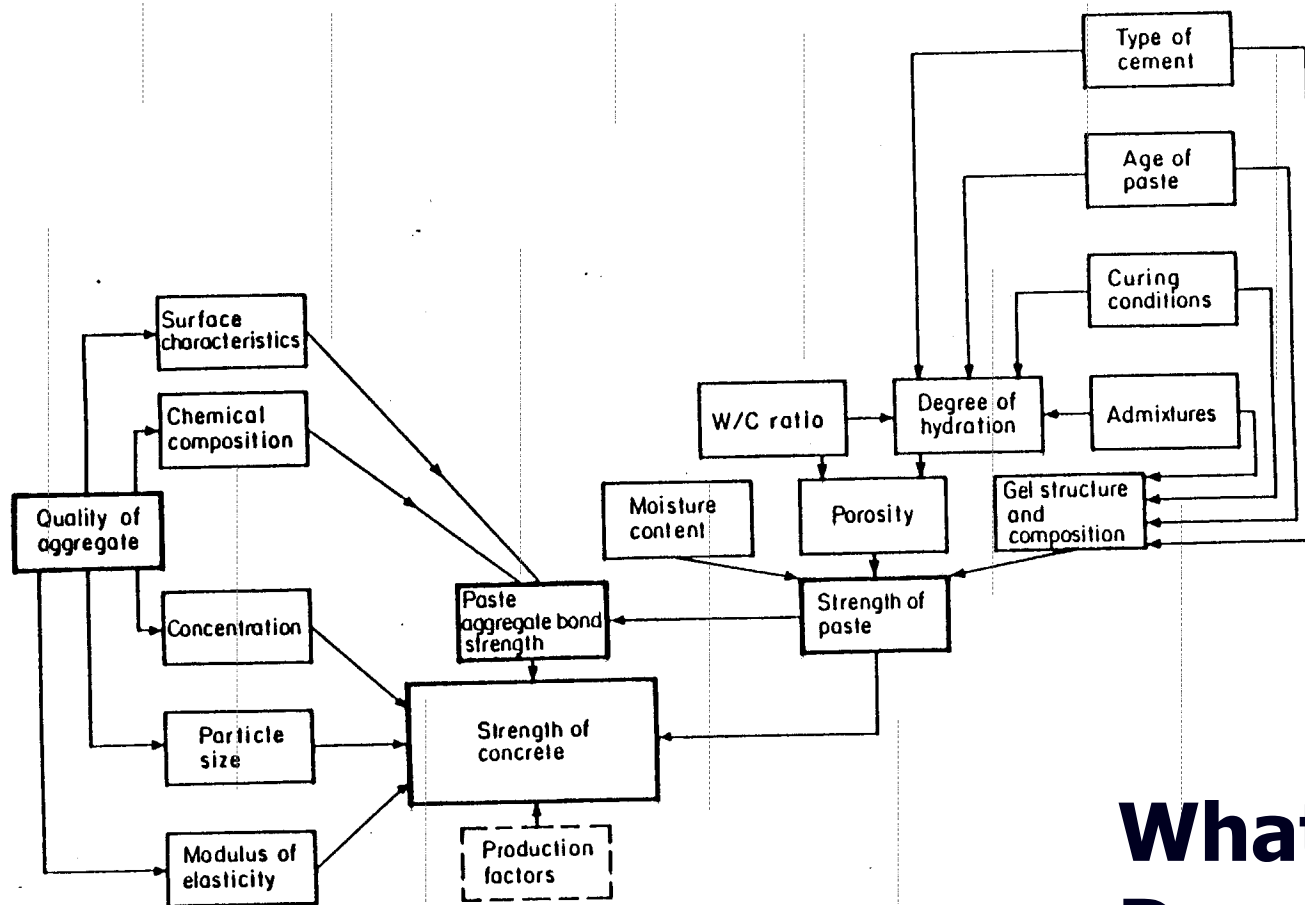
- The principal properties of hardened concrete which are of practical importance can be listed as:
 - Strength
 - Permeability & durability
 - Shrinkage & creep deformations
 - Response to temperature variations
- Of these compressive strength is the most important property of concrete. Because;

PROPERTIES OF HARDENED CONCRETE

- Of the abovementioned hardened properties compressive strength is one of the most important property that is often required, simply because;
 - Concrete is used for compressive loads
 - Compressive strength is easily obtained
 - It is a good measure of all the other properties.



What Affects Concrete Strength



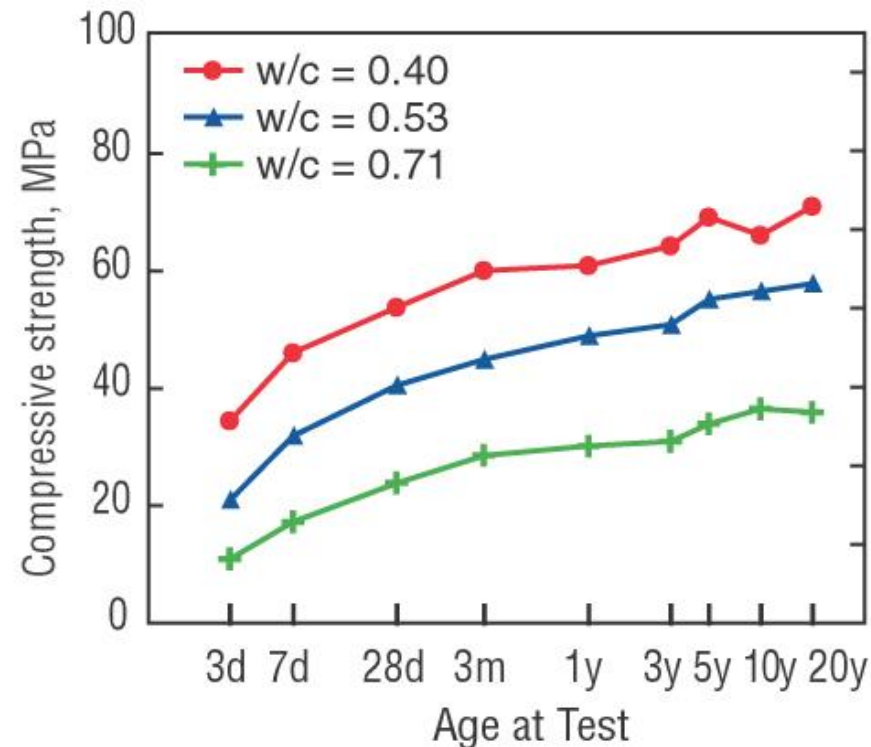
**What
Doesn't?**

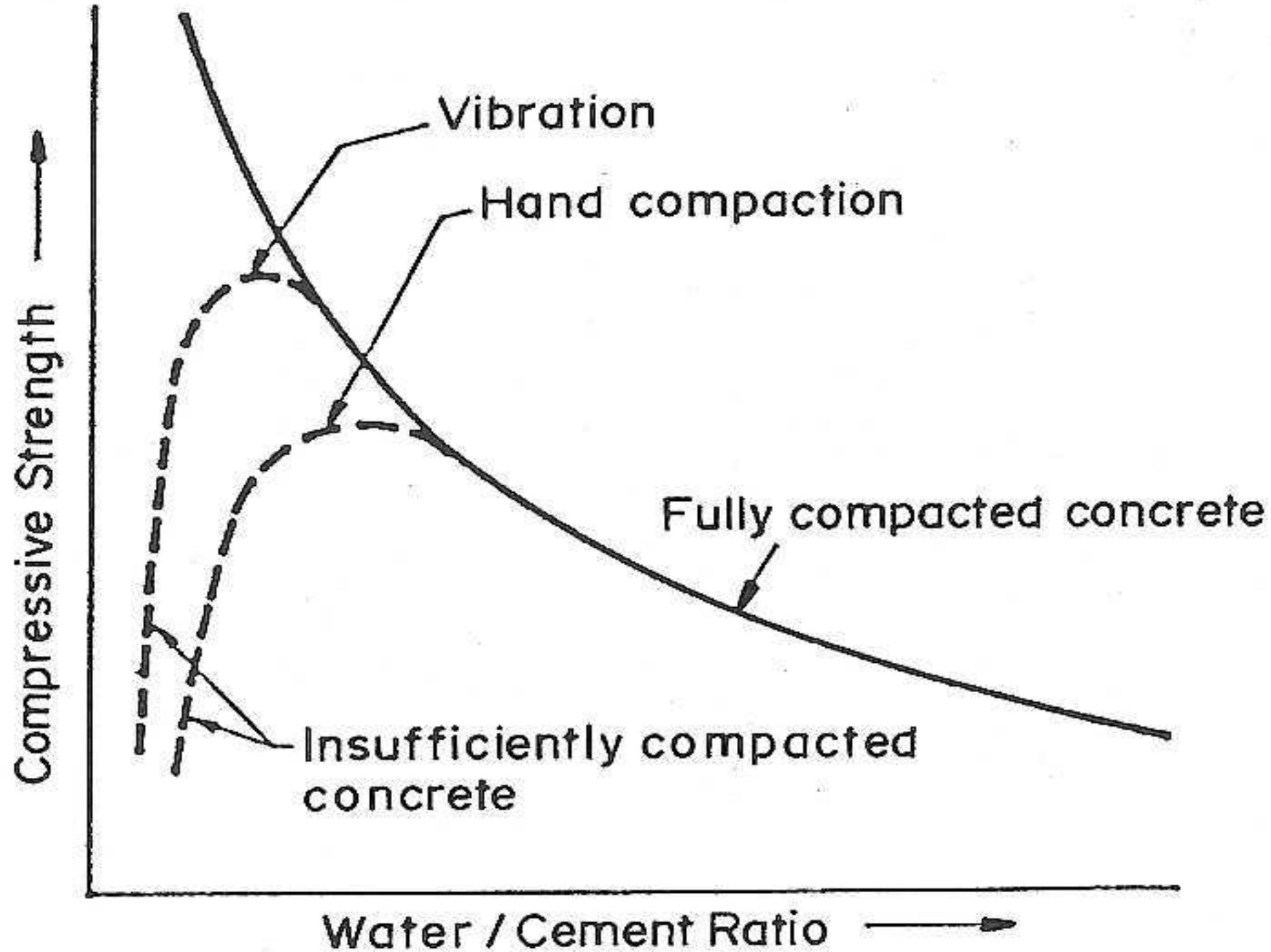
Factors Affecting Strength

- Effect of materials and mix proportions
- Production methods
- Testing parameters

STRENGTH OF CONCRETE

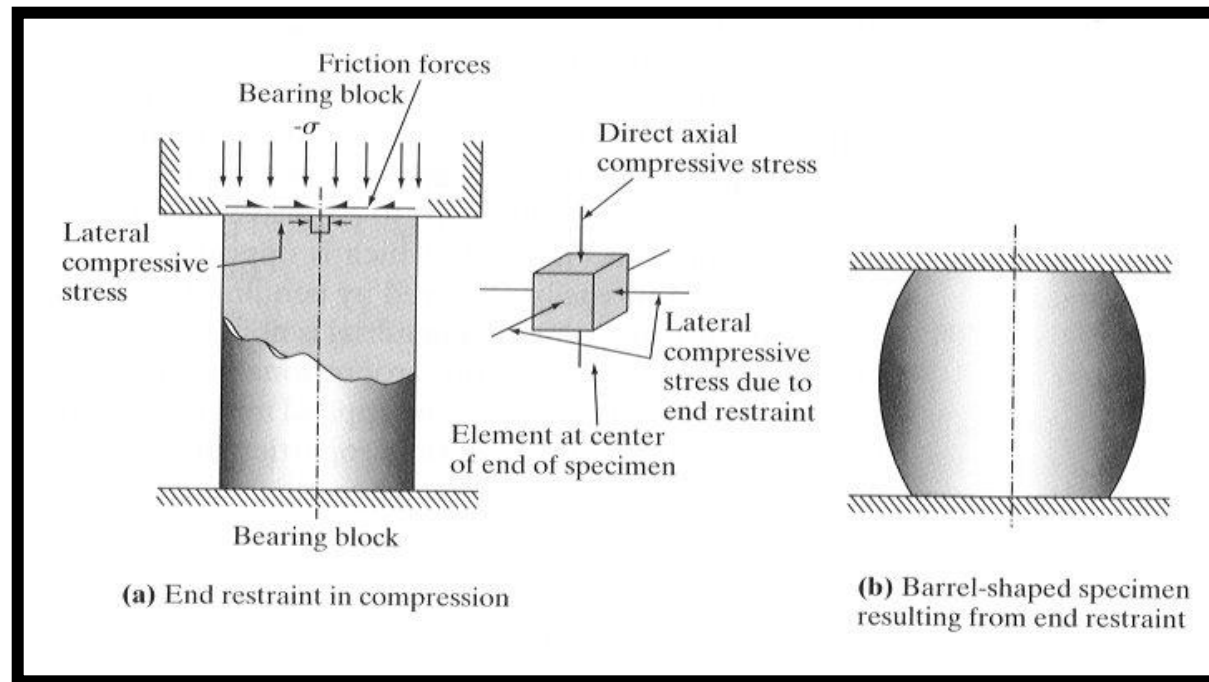
- The strength of a concrete specimen prepared, cured and tested under specified conditions at a given age depends on:
 - w/c ratio
 - Degree of compaction





COMPRESSIVE STRENGTH

- Compressive Strength is determined by loading properly prepared and cured cubic, cylindrical or prismatic specimens under compression.



COMPRESSIVE STRENGTH

- Cubic: 15x15x15 cm
- Cubic specimens are crushed after rotating them 90° to decrease the amount of friction caused by the rough finishing.
- Cylinder: $h/D=2$ with $h=15$
- To decrease the amount of friction, capping of the rough casting surface is performed.

COMPRESSIVE STRENGTH



Cubic specimens
without capping



**Cylindrical specimens
with capping**

COMPRESSIVE STRENGTH



Bonded sulphur capping



Unbonded neoprene pads

STRENGTH CLASSES (TS EN 206-1)

- The compressive strength value depends on the shape and size of the specimen.

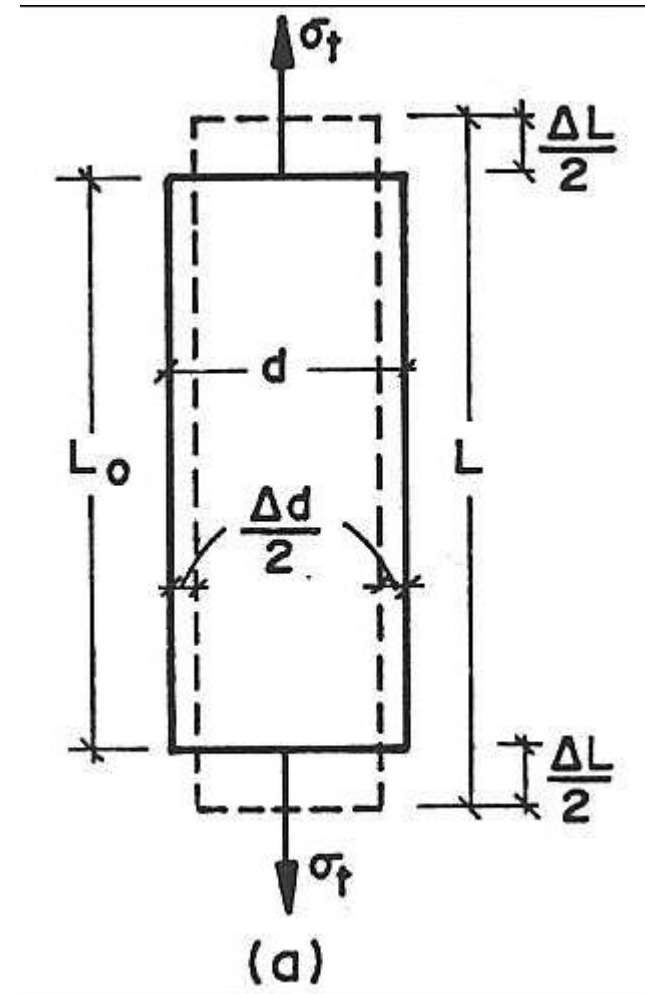
Basınç dayanımı sınıfı	En düşük karakteristik silindir dayanımı	En düşük karakteristik küp dayanımı
	$f_{ck,sil}$ N/mm ²	$f_{ck,küp}$ N/mm ²
C 8/10	8	10
C 12/15	12	15
C 16/20	16	20
C 20/25	20	25
C 25/30	25	30
C 30/37	30	37
C 35/45	35	45
C 40/50	40	50
C 45/55	45	55
C 50/60	50	60
C 55/67	55	67
C 60/75	60	75
C 70/85	70	85
C 80/95	80	95
C 90/105	90	105
C 100/115	100	115

TENSILE STRENGTH

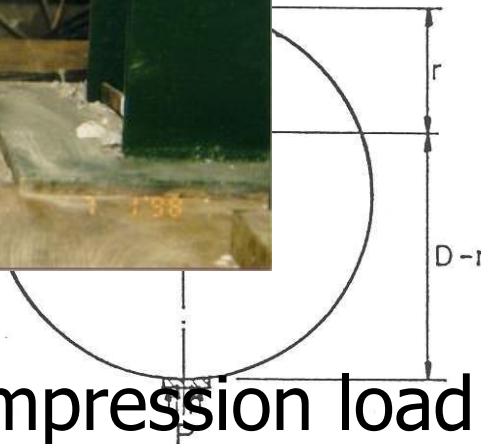
- Tensile Strength can be obtained either by direct methods or indirect methods.

Direct methods suffer from a number of difficulties related to holding the specimen properly in the testing machine without introducing stress concentration and to the application of load without eccentricity.

DIRECT TENSILE STRENGTH



SPLIT TENSILE STRENGTH



Due to applied compression load a fairly uniform tensile stress is induced over nearly $2/3$ of the diameter of the cylinder perpendicular to the direction of load application.

$$\sigma_{st} = \frac{2P}{\pi DI}$$

P: applied compressive load

D: diameter of specimen

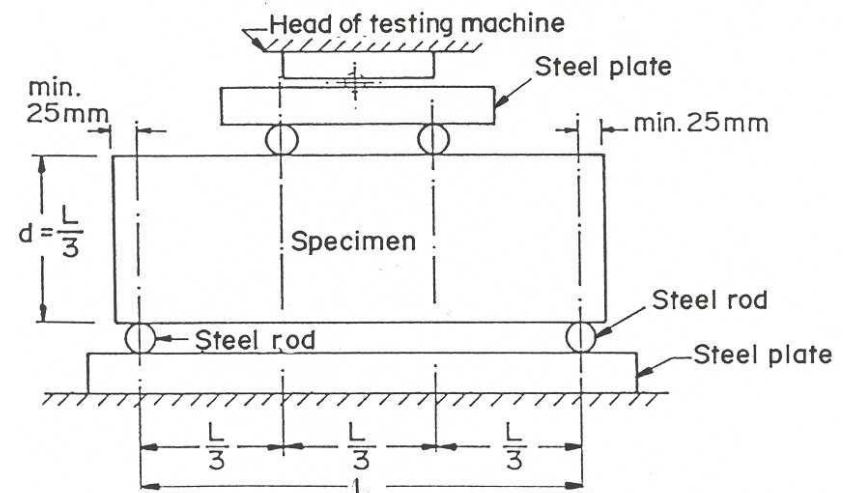
- Splitting Tensile Strength
The advantage of the splitting test over the direct tensile test is the same molds are used for compressive & tensile strength determination.

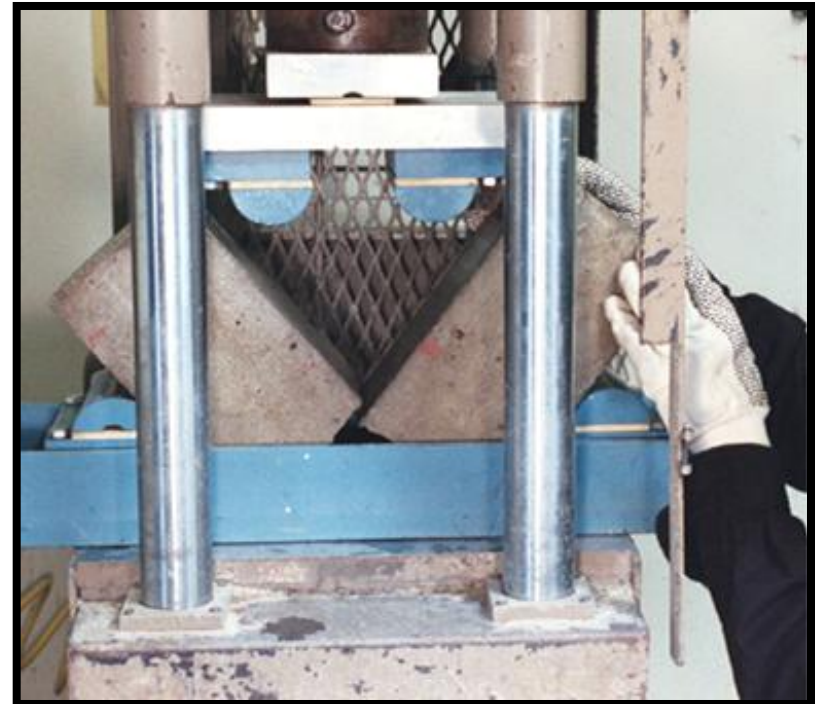
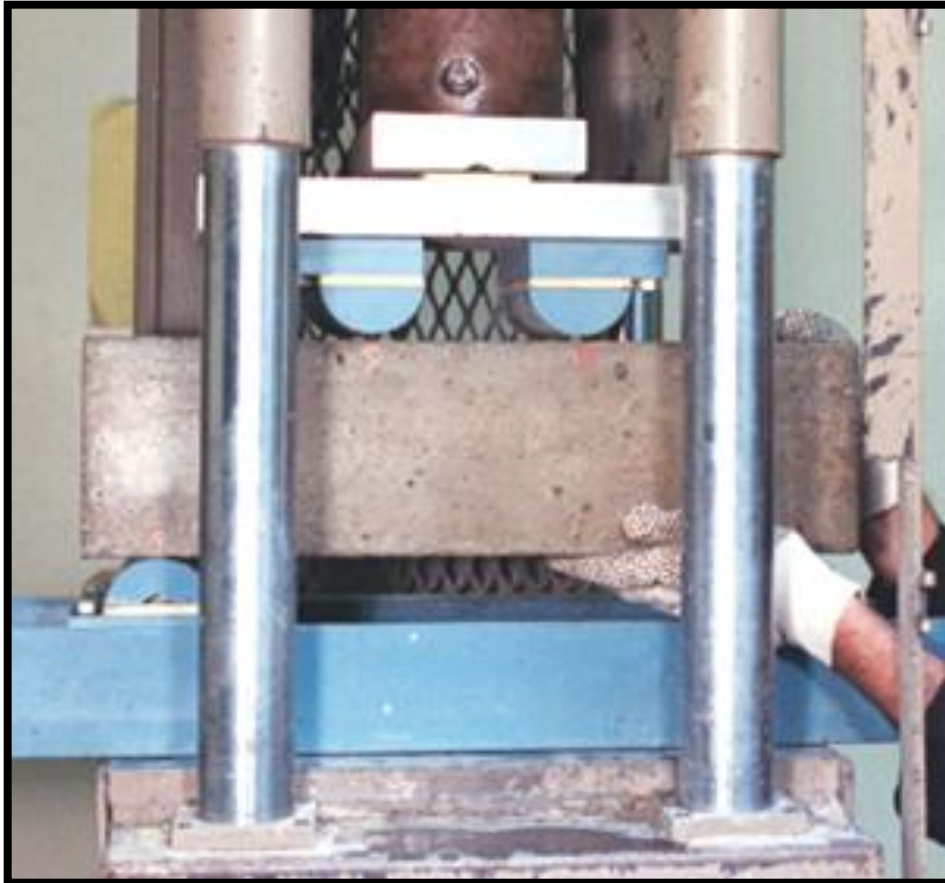
- The test is simple to perform and gives uniform results than other tension tests.

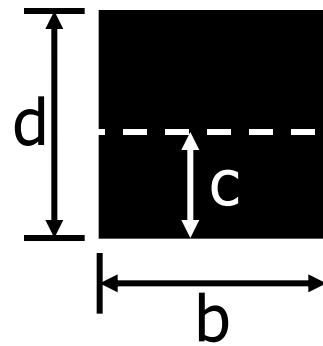
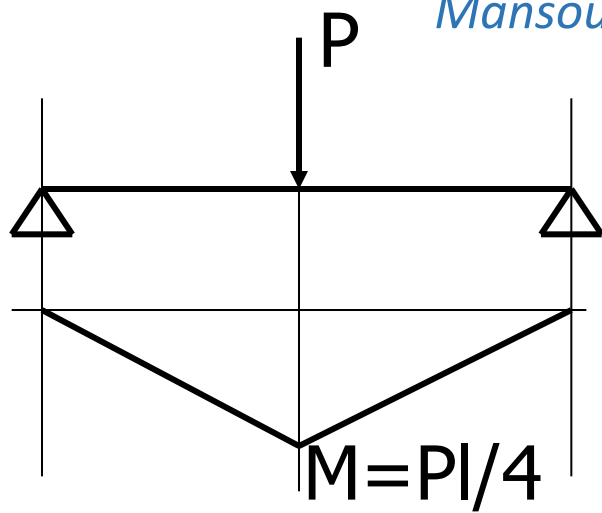
FLEXURAL STRENGTH

- The flexural tensile strength at failure or the modulus of rupture is determined by loading a prismatic concrete beam specimen.

The results obtained are useful because concrete is subjected to flexural loads more often than it is subjected to tensile loads.

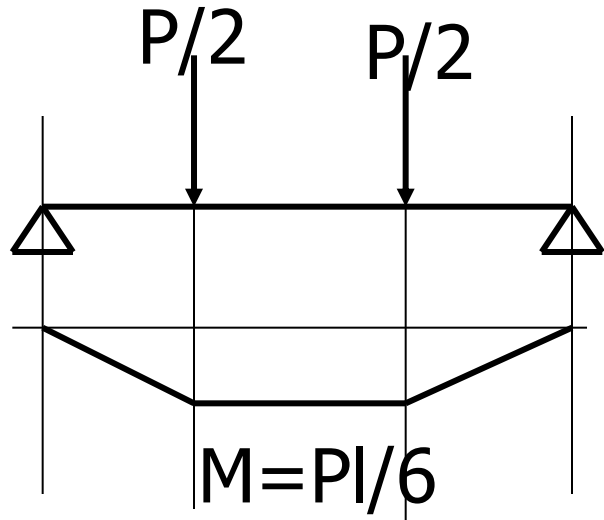






$$I = \frac{bd^3}{12}$$

$$\sigma = \frac{M c}{I} = \frac{(Pl/4) (d/2)}{bd^3/12} = \frac{3}{2} \frac{Pl}{bd^2}$$



$$\sigma = \frac{(Pl/6) (d/2)}{bd^3/12} = \frac{Pl}{bd^2}$$

Factors Affecting the Strength of Concrete

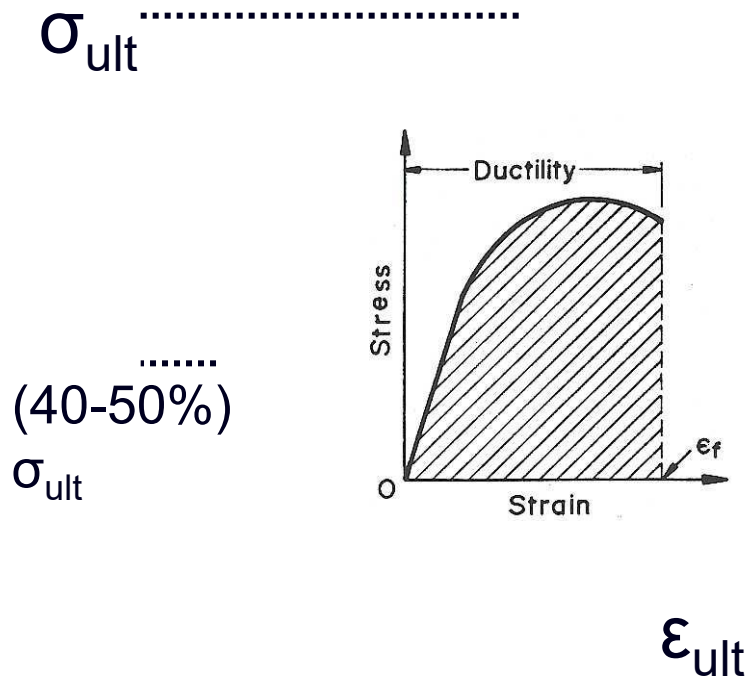
- Factors depended on the **2.** test type:

- Size of specimen
- Size of specimen in relation with size of agg.
- Support condition of specimen
- Moisture condition of specimen
- Type of loading adopted
- Rate of loading
- Type of test machine

- **Factors independent of test type:**

- Type of cement
- Type of agg.
- Degree of compaction
- Mix proportions
- Type of curing
- Type of stress situation

STRESS-STRAIN RELATIONS IN CONCRETE

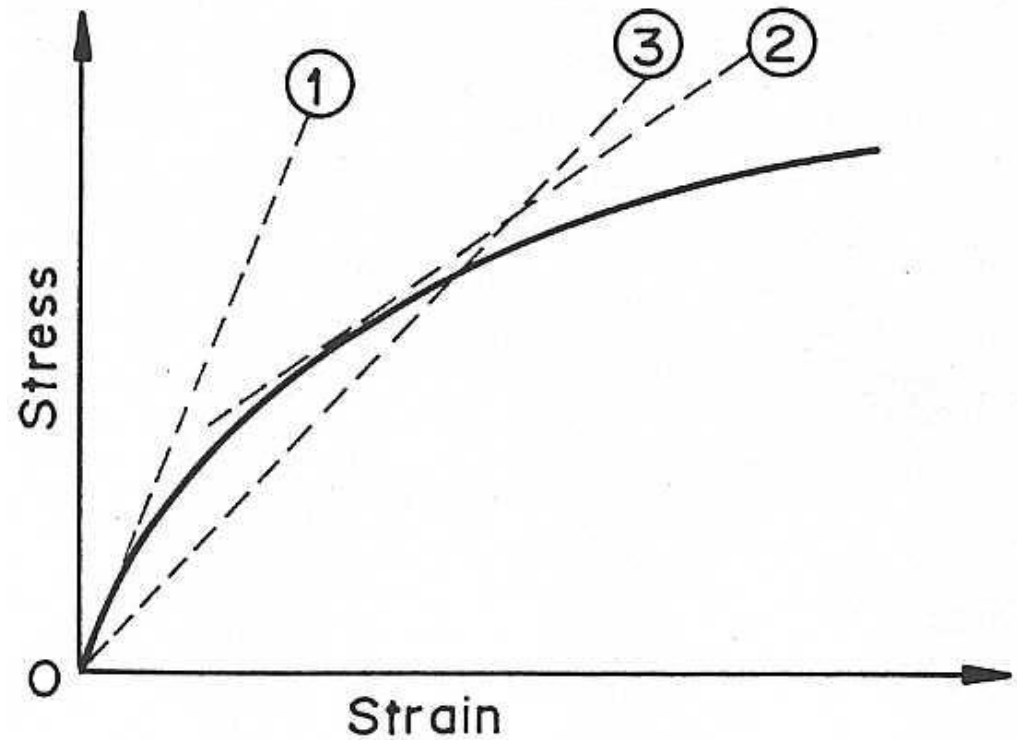


σ - ϵ relationship for concrete is nonlinear. However, specially for cylindrical specimens with $h/D=2$, it can be assumed as linear upto 40-50% of σ_{ult}

MODULUS OF ELASTICITY OF CONCRETE

- Due to the nonlinearity of the σ - ϵ diagram, E is defined by:

- Initial Tangent Method
- Tangent Method
- Secant Method



ACI $\rightarrow E=15200 \sigma_{ult}^{1/2} \rightarrow$ 28-D cylindrical comp.str. (kgf/cm²)

TS $\rightarrow E=15500 W^{1/2} \rightarrow$ 28-D cubic comp.str. (kgf/cm²)